Unit-1 Determinant & Function - 16 Marks

* Determinant

$a_1 x + b_1 y = c_1$ $\qquad$ $2x + 3y = 1$

$a_2 x + b_2 y = c_2$ $\qquad$ $x - y = 4$

$$\frac{x}{\begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix}} = \frac{1}{\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix}}$$

* 2×2 Determinant

→ Leading element

$$\begin{array}{ccc} & & \\ D = & \begin{vmatrix} \boxed{a_1} & b_1 \\ a_2 & b_2 \end{vmatrix} & \begin{array}{l} \to R_1 \\ \to R_2 \end{array} \\ & \begin{array}{cc} \downarrow & \downarrow \\ C_1 & C_2 \end{array} & \end{array}$$

Subsidiary diagonal $\qquad$ Principal diagonal

* Solution of 2×2 Determinant

$$* \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - b_1 a_2$$

2×2 is order of Determinant.
↑ Row ↑ column

* why determinant is always square type ?

2×2, 3×3, 4×4 ....

* Solve the following Determinents.

* $\begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} 2 & -3 \\ 5 & 4 \end{vmatrix}$

* $\begin{vmatrix} 5 & -8 \\ 9 & 2 \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} 5 & 4 \\ 1 & -2 \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} -3 & 1 \\ -4 & 2 \end{vmatrix}$

* $\begin{vmatrix} -2 & 5 \\ -3 & 2 \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{vmatrix}$ $\qquad$ * $\begin{vmatrix} \sec\theta & \tan\theta \\ \tan\theta & \sec\theta \end{vmatrix}$

* If $\begin{vmatrix} x & 1 \\ 4 & 2 \end{vmatrix} = 0$ then $x =$ ______

* If $\begin{vmatrix} x & 3 \\ -2 & 2 \end{vmatrix} = 2$ then $x =$ ______

* If $\begin{vmatrix} x & -3 \\ y & 3 \end{vmatrix} = 9$ then $x + y =$ ______

* If $\begin{vmatrix} 2 & x \\ -3 & 5 \end{vmatrix} = 13$ then $x =$ ______

* If $\begin{vmatrix} x-5 & 4 \\ 6 & x+5 \end{vmatrix} = 0$ then $x =$ ____.

* If $\begin{vmatrix} x-1 & 2 \\ 4 & x+1 \end{vmatrix} = 0$ then $x =$ ____.

* 3X3 Determinent

$2x - y + z = 1$

$x + 3y - 2z = 2$

$2x - y + 4z = 3$

$\Rightarrow \dfrac{x}{|\quad|} = \dfrac{-y}{|\quad|} = \dfrac{z}{|\quad|} = \dfrac{1}{|\quad|}$

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

$3 \times 3 \Rightarrow$ order of Determinent

Row column

* solution of 3X3 Determinent (by Row or column method)

$$D = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2 c_3 - c_2 b_3) - b_1(a_2 c_3 - c_2 a_3) + c_1(a_2 b_3 - b_2 a_3)$$

* $D = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 0 & -2 \\ 4 & 2 & 1 \end{vmatrix} = (-12)$

* $D = \begin{vmatrix} 3 & 7 & 3 \\ 2 & -5 & 2 \\ 1 & 8 & 1 \end{vmatrix} = (0)$

* $D = \begin{vmatrix} 2 & 6 & -3 \\ 1 & 2 & 0 \\ 3 & 4 & -2 \end{vmatrix} = (10)$

* $D = \begin{vmatrix} 3 & -2 & 1 \\ 1 & -3 & 2 \\ -5 & 4 & 5 \end{vmatrix} = (-50)$

* $D = \begin{vmatrix} 3 & -1 & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 0 \end{vmatrix} = (2)$

* $D = \begin{vmatrix} 3 & -2 & -2 \\ 1 & 4 & 0 \\ -2 & 3 & -1 \end{vmatrix} = (-36)$

* $D = \begin{vmatrix} 1 & 7 & -3 \\ -4 & 6 & 2 \\ 2 & -5 & 3 \end{vmatrix} = (116)$

* $\begin{vmatrix} 1 & 1 & 2 \\ 3 & 5 & -1 \\ 2 & 2 & 4 \end{vmatrix} (R_1 = R_3)$ * $\begin{vmatrix} 2 & -2 & 1 \\ 4 & 7 & 2 \\ 6 & 0 & 3 \end{vmatrix} (C_1 = C_3)$ * $\begin{vmatrix} a & b & c \\ x & y & z \\ a & b & c \end{vmatrix} (R_1 = R_3)$

* any two row or any two column are same then answer of Determinant is zero.

* solve $\begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix}$

$$= \begin{vmatrix} a+b+c & b+c & 1 \\ b+c+a & c+a & 1 \\ c+a+b & a+b & 1 \end{vmatrix} \quad (C_1+C_2$$

$$= (a+b+c)\begin{vmatrix} 1 & b+c & 1 \\ 1 & c+a & 1 \\ 1 & a+b & 1 \end{vmatrix}$$

$$= (a+b+c)(0) \quad (\because C_1 = C_3)$$

$$= 0$$

* solve $\begin{vmatrix} x & x+a & a \\ y & y+b & b \\ z & z+c & c \end{vmatrix}$

$$= \begin{vmatrix} x & x+a & x+a \\ y & y+b & y+b \\ z & z+c & z+c \end{vmatrix} \quad (C_3+C_1)$$

$$= 0 \quad (C_2 = C_3)$$

* If $\begin{vmatrix} x-2 & 2 & 2 \\ -1 & x & -2 \\ 2 & 0 & 4 \end{vmatrix} = 0$ then find value of x.

$(x = 0$ OR $x = 3)$

* Sarru's Method

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{matrix} a_1 & \boxed{b_1} \\ a_2 & b_2 \\ a_3 & b_3 \end{matrix}$$

$$= [a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3] - [b_1 a_2 c_3 + a_1 c_2 b_3 + c_1 b_2 a_3]$$

* $D = \begin{vmatrix} 2 & 3 & -1 \\ 4 & 0 & 5 \\ -3 & 2 & 4 \end{vmatrix} = (-121)$

* $D = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (0)$

* Function

* Let A and B be two non-empty sets. If every element of A is related to a unique element of B, then the relation is called the Function from A to B.

→ It is denoted as $f: A \to B, \forall x \in A \ \exists$ a unique element of B, say y, such that

$$\boxed{y = f(x)}$$

→ $f : \underset{\substack{\uparrow \\ \text{Domain}}}{A} \to B.$ ← Co-Domain.

→ $y = f(x)$ is the relation for independent variable $x \in A$ and dependent variable $x \in B$.

* $f : A \to B$, $f(x) = x + 1$. find $f(1)$

$f(1) = 1 + 1 = 2$

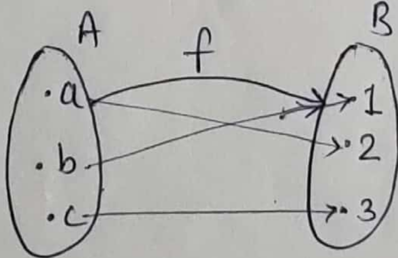
* $f : A \to B$ $f(x) = x + 1$

$A = \{1, 2, 3\}$ $B = \{2, 3, 4, 5, 6\}$

$f(1) = 2$

$f(2) = 3$

$f(3) = 4$



$x_1 \neq x_2$
$\Rightarrow f(x_1) \neq f(x_2)$

* $f(1) = 2$
$f(a) = 2$
$\Rightarrow f(1) = f(a)$
$\therefore a = 1$

$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$

* Types of Function

* one - one function $\Rightarrow$ $f(x_1) = f(x_2)$

$\Rightarrow x_1 = x_2$

* Many - one function $\Rightarrow$ It function is not one-one then its called many-one function.

* odd function : $\Rightarrow$ $f(-x) = -f(x)$.

$f(x) = x^3$

$f(-x) = (-x)^3 = -x^3 = -f(x)$

x^3, is an odd function.

* even function $\Rightarrow$ $f(-x) = f(x)$

$f(x) = x^2$

$f(-x) = (-x)^2 = x^2 = f(x)$

x^2 is an even function.

* If $f(x) = \log x$ then $f(1) =$ ____
* If $f(x) = \log x$ then $f(x) - f(y) =$ ____
* If $f(x) = \log x$ then $f(x) + f(y) =$ ____
* If $f(x) = \log \frac{1}{x}$ then $f(100) =$ ____
* If $f(x) = x^2 - 1$ then $f(-1) =$ ____
* If $f(x) = x^3 - 1$ then $f(2) + f(-3) =$ ____
* If $f(x) = 2^x - x^2$ then $f(2) =$ ____
* If $f(x) = 2^x - \log_2 x$ then $f(2) =$ ____
* If $f(x) = \log_2 x$ then $f(4) =$ ____
* If $f(x) = \log(e^x)$ then $f(0) =$ ____
* If $f(x) = \log(e^x)$ then $f(-1) =$ ____
* If $f(x) = \frac{a^x + a^{-x}}{2}$ is ____ function. (odd, even)

* If $f(x) = \log x$ then prove that (i) $f(xy) = f(x) + f(y)$
(ii) $f\left(\frac{x}{y}\right) = f(x) - f(y)$ (iii) $f(x^2) = 2f(x)$.

* If $f(x) = e^x$ then prove that (i) $f(x).f(y) = f(x+y)$
(ii) $\frac{f(x)}{f(y)} = f(x-y)$.

* If $f(x) = a^x$ then prove that (i) $f(x+y) = f(x).f(y)$
(ii) $f(x-y) = \frac{f(x)}{f(y)}$.

* If $f(x) = a^x$ then prove that $f(x+1) - f(x) = (a-1)f(x)$

* If $f(x) = 4^x$ then prove that $f(x+1) - f(x) = 3f(x)$.

* If $f(x) = \frac{1}{x+1}$ then prove that $f(x) + f\left(\frac{1}{x}\right) = 1$

* If $f(x) = \frac{ax+b}{bx+a}$ then show that $f(x).f\left(\frac{1}{x}\right) = 1$.

* If $f(x) = \frac{1-x}{1+x}$ then P. T. (i) $f(x) + f\left(\frac{1}{x}\right) = 0$
(ii) $f(x) - f\left(\frac{1}{x}\right) = 2.f(x)$ (iii) $f(x).f(-x) = 1$.

* If $f(x) = \log\left(\frac{x}{x-1}\right)$ then P. T. $f(a+1) + f(a) = \log\left(\frac{a+1}{a-1}\right)$

* If $f(x) = \log\left(\frac{x-1}{x}\right)$ then P. T. $f(x) + f(-x) = f(x^2)$.

* If $f(x) = \log\left(\frac{1+x}{1-x}\right)$ then P. T. $f(x) + f(-x) = 0$

* If $f(x) = \log\left(\frac{1-x}{1+x}\right)$ then P. T. $f\left(\frac{2x}{1+x^2}\right) = 2.f(x)$.

* If $f(x) = \frac{1+x}{1-x}$ then P. T. $f\left(\frac{x+y}{1+xy}\right) = f(x).f(y)$.

* If $f(x) = \log_2 x$ and $g(x) = x^4$ then find $f(g(2))$.

* If $f(x) = \frac{1+x}{1-x}$ then P. T. $x f(f(x)) + 1 = 0$

* If $f(x) = \tan x$ then prove that

(i) $f(x+y) = \dfrac{f(x)+f(y)}{1-f(x)f(y)}$

(ii) $f(2x) = \dfrac{2f(x)}{1-(f(x))^2}$

* If $f(x) = \sin x$ then show that $2f(x) \cdot f(\frac{\pi}{2}+x) = f(2x)$.

* If $f(x) = \dfrac{x+3}{4x-5}$ and $t = \dfrac{3+5x}{4x-1}$ then prove that $x = f(t)$.

* Index & Indices Rules

- $\rightarrow a^{x} \cdot a^{y} = a^{x+y}$
- $\rightarrow \dfrac{a^{x}}{a^{y}} = a^{x-y}$
- $\rightarrow (a^{m})^{n} = a^{m \cdot n}$

- $\rightarrow \log x + \log y = \log (xy)$
- $\rightarrow \log x - \log y = \log \left(\dfrac{x}{y}\right)$
- $\rightarrow \log x^{n} = n \cdot \log x$
- $\rightarrow \log_{y} x = \dfrac{\log x}{\log y}$

* Values of Logarithm

$\log 1 = 0$, $\log_{a} a = 1$

$a^{\log_{a} y} = y$

$* \; a^{x} = n \iff \log_{a} n = x$

Exponential form — Logarithmic form.

- $* \; 2^{3} = 8$
- $* \; 3^{4} = 81$
- $* \; 9^{3/2} = 27$
- $* \; 10^{-2} = 0.01$
- $* \; (-2)^{3} = -8$

- $* \; \log_{2} 32 = 5$
- $* \; \log_{5} 125 = 3$
- $* \; \log_{10} 0.001 = 3$
- $* \; \log_{12} \dfrac{1}{144} = 2$

* $\log_{15} 1 =$ ___
* $\log 1 \cdot \log 2 \cdot \log 3 \cdots \log 2000 =$ ___
* $\log_2 8 =$ ___
* $\log_2 64 =$ ___
* $\log_5 125 =$ ___
* $\log_a \frac{1}{a} =$ ___
* $\log_3 \frac{1}{27} =$ ___
* $\log_2 \frac{1}{8} =$ ___
* $\log_{10} 0.001 =$ ___
* $\log_b a \times \log_a b =$ ___
* $\log(\tan\theta) + \log(\cot\theta) =$ ___
* $\log 2 + \log \frac{1}{2} =$ ___
* $\log a + \log \frac{1}{a} =$ ___
* $\log m - \log n =$ ___
* $\log 32 \div \log 16 =$ ___
* $\log 27 \div \log 9 =$ ___
* $\log 32 - \log 16 =$ ___
* If $\log_7 x = 1$ then $x =$ ___
* If $\log_a 32 = 5$ then $a =$ ___

* $a^{\log_a b} =$ ___
* $2^{-\log_2 3} =$ ___
* $1024^{\log_2 m} =$ ___
* If $\log x + \log 2x = \log 18$ then $x =$ ___
* $\log_{10}(x+1) + \log_{10}(x-1) = \log_{10} 3$ then $x =$ ___
* $2^{\log_4 9}$
* $9^{\log_3 2}$
* $16^{\log_4 5}$
* $1024^{\log_2 m}$
* $\sqrt{3}^{\log_3 81}$

* $\log\left(\frac{x^2}{yz}\right) + \log\left(\frac{y^2}{zx}\right) + \log\left(\frac{z^2}{xy}\right)$.

* $\log\left(\frac{a-b}{b-c}\right) + \log\left(\frac{b-c}{c-a}\right) + \log\left(\frac{c-a}{a-b}\right)$.

* $\log\left(\frac{25}{32}\right) + \log\left(\frac{225}{64}\right) + \log\left(\frac{128}{25}\right) + \log\left(\frac{32}{450}\right)$.

* $\log\left(\frac{9}{14}\right) - \log\left(\frac{15}{16}\right) + \log\left(\frac{35}{24}\right)$.

* P.T. $\log\left(\frac{75}{16}\right) - 2\log\left(\frac{5}{9}\right) + \log\left(\frac{32}{243}\right) = \log 2$

* P.T. $\log(x+\sqrt{x^2-1}) + \log(x-\sqrt{x^2-1}) = 0$.

* P.T. $\log(\sqrt{x^2+1}+x) + \log(\sqrt{x^2+1}-x) = 0$.

P.T. $2\log\left(\frac{6}{7}\right) + \frac{1}{2}\log\left(\frac{81}{16}\right) - \log\left(\frac{27}{196}\right) = \log 12$

Simplify $\log 2 + 16\log\left(\frac{16}{15}\right) + 12\log\left(\frac{25}{24}\right) + 7\log\left(\frac{81}{80}\right)$.

* Change of base

* P.T. $\frac{1}{\log_3 6} + \frac{1}{\log_2 6} = 1$

* P.T. $\frac{1}{\log_5 15} + \frac{1}{\log_3 15} = 1$

* P.T. $\frac{1}{\log_{24} 12} + \frac{1}{\log_8 12} + \frac{1}{\log_9 12} = 3$

* P.T. $\frac{1}{\log_6 24} + \frac{1}{\log_{12} 24} + \frac{1}{\log_8 24} = 2$

* P.T. $\frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc} = 1$

* P.T. $\frac{1}{\log_{12} 60} + \frac{1}{\log_{30} 60} + \frac{1}{\log_{10} 60} = 2$

* P.T. $\frac{1}{\log_{xy} xyz} + \frac{1}{\log_{yz} xyz} + \frac{1}{\log_{zx} xyz} = 2$

* P.T. $\log_m x + \log_{m^2} x^2 + \log_{m^3} x^3 + \log_{m^4} x^4 = 4 \cdot \log_m x$

* P.T. $\log_b a \cdot \log_c b \cdot \log_a c = 1$

* P.T. $\log_x y \cdot \log_y z \cdot \log_z \omega = \log_x \omega$

* $\log_y x^2 \cdot \log_z y^3 \cdot \log_x z^4 = 24$

* $\log_{\sqrt{q}} p^2 \cdot \log_{\sqrt{r}} q^2 \cdot \log_{\sqrt{p}} r^2 = 64$

* $\log_{b^3} a^2 \cdot \log_{a^3} b^2 \cdot \log_{a^3} c^2 = \frac{8}{27}$

* $\frac{1}{\log_x yz + 1} + \frac{1}{\log_y zx + 1} + \frac{1}{\log_z xy + 1} = 1$

* If $\log\left(\frac{a+b}{2}\right) = \frac{1}{2}(\log a + \log b)$ then P.T. $a^2+b^2 = 2ab$.
* If $\log\left(\frac{a+b}{2}\right) = \frac{1}{2}(\log a + \log b)$ then P.T. $a = b$.
* If $\log\left(\frac{x+y}{3}\right) = \frac{1}{2}(\log x + \log y)$ then P.T. $x^2+y^2 = 7xy$
* If $\log\left(\frac{a-b}{2}\right) = \frac{1}{2}(\log a + \log b)$ then P.T. $a^2+b^2 = 6ab$

  OR $\frac{a}{b} + \frac{b}{a} = 6$.

* If $\log\left(\frac{x+y}{3}\right) = \frac{1}{2}(\log x + \log y)$ then P.T. $\frac{x}{y} + \frac{y}{x} = 7$.
* If $\frac{4\log 3 \times \log x}{\log 9} = \log 27$ then find the value of x.
* If $\frac{\log x \times \log 16}{\log 32} = \log 256$ then find the value of x.
* If $\frac{2\log_5^x + \log_5^9}{\log_5 3x} = \log_5^x$ then find the value of x.
* Solve $\log x^3 - \log 25 = \log x$
* Simplify $\log_3 84 - \log_3 28 - 3^{\log_3 1}$
* Solve $\log x + \log(x-5) = \log 6$
* Solve $\log_2(x+5) + \log_2(x-2) = 3$
* Solve $\log_2(\log_3(\log_2^x)) = 1$.

$\log\left(\frac{1}{x^2} + \frac{1}{y^2}\right) = \log 2 - \log x + \log\left(\frac{1}{y}\right)$ then P.T.

$x = y$

Unit-2. Trigonometry - 14 Marks

* **Unit circle**

$x^2 + y^2 = 1$

* convert Degree into Radian form

* 60° * 135° * 75° * 30° * 45°

* 330° * 105°

* convert Radian into Degree form

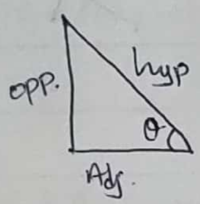
* $\frac{\pi}{15}$ * $\frac{\pi}{30}$ * $\frac{11\pi}{6}$ * $\frac{7\pi}{4}$

* $\frac{7\pi}{6}$ * $\frac{\pi}{12}$ * $\frac{2\pi}{9}$ * $\frac{3\pi}{20}$

* $\sin\theta = \frac{\text{Opp. S.}}{\text{hyp.}}$

* $\cos\theta = \frac{\text{Adj. S.}}{\text{hyp}}$

* $\tan\theta = \frac{\text{Opp. S}}{\text{Adj. S.}}$



* $\sin\theta = \frac{1}{\operatorname{cosec}\theta}$
* $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$
* $\cos\theta = \frac{1}{\sec\theta}$
* $\sec\theta = \frac{1}{\cos\theta}$
* $\tan\theta = \frac{\sin\theta}{\cos\theta}$
* $\cot\theta = \frac{\cos\theta}{\sin\theta}$
* $\sin\theta \cdot \operatorname{cosec}\theta = 1$
* $\cos\theta \cdot \sec\theta = 1$
* $\tan\theta \cdot \cot\theta = 1$

* $\sin^2\theta + \cos^2\theta = 1$
* $\sec^2\theta - \tan^2\theta = 1$
* $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

*

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* P.T. $\dfrac{\sin(\pi+\theta)}{\sin(2\pi-\theta)} + \dfrac{\tan(\frac{\pi}{2}+\theta)}{\cot(\pi-\theta)} + \dfrac{\cos(2\pi+\theta)}{\sin(\frac{\pi}{2}+\theta)} = 3$

* P.T. $\dfrac{\sin(\frac{\pi}{2}-\theta)}{\cos(\pi-\theta)} + \dfrac{\tan(\frac{\pi}{2}+\theta)}{\cot(\pi+\theta)} + \dfrac{\operatorname{cosec}(\frac{\pi}{2}+\theta)}{\sec(\pi+\theta)} = -3.$

* Evaluate $\dfrac{\sin(\theta-\frac{3\pi}{2})}{\cos(\theta-2\pi)} + \dfrac{\sec(\frac{3\pi}{2}+\theta)}{\operatorname{cosec}(\pi+\theta)} + \dfrac{\cot(\frac{\pi}{2}+\theta)}{\tan(2\pi+\theta)}$ (-1)

* Evaluate $\dfrac{\sin(\theta-\frac{\pi}{2})}{\cos(\theta-\pi)} + \dfrac{\tan(\frac{\pi}{2}+\theta)}{\cot(\pi+\theta)} + \dfrac{\operatorname{cosec}(\frac{\pi}{2}+\theta)}{\sec(\pi+\theta)}$ (-1)

* P.T. $\dfrac{\sin(\pi-A)}{\tan(\pi+A)} \cdot \dfrac{\cot(\frac{\pi}{2}-A)}{\tan(\frac{\pi}{2}+A)} \cdot \dfrac{\cos(2\pi-A)}{\sin(-A)} = \sin A$

* Simplify $\dfrac{\sin(180^\circ-\theta)\cos(270-\theta)\operatorname{cosec}(90^\circ+\theta)}{\sec(270^\circ+\theta)\cot(90^\circ+\theta)\tan(360^\circ+\theta)}$ $(\sin\theta\cos\theta)$

* P.T. $\cos\frac{19\pi}{6}\cdot\sin\frac{17\pi}{6} - \sin\frac{11\pi}{6}\cos\frac{13\pi}{6} = 0$

* P.T. $\sin^2\frac{\pi}{4} + \sin^2\frac{3\pi}{4} + \sin^2\frac{5\pi}{4} + \sin^2\frac{7\pi}{4} = 2$

* P.T. $\tan\frac{\pi}{20}\tan\frac{3\pi}{20}\tan\frac{5\pi}{20}\tan\frac{7\pi}{20}\tan\frac{9\pi}{20} = 1$

* P.T. $\cot\frac{\pi}{20}\cot\frac{3\pi}{20}\cot\frac{5\pi}{20}\cot\frac{7\pi}{20}\cot\frac{9\pi}{20} = 1$

* P.T. $\sin 407\cdot\sin 133 + \sin 317\cdot\sin 223 = 1$

* P.T. $\tan 225\cdot\cot 405 + \tan 485\cdot\cot 315 = 0$

* FOR ΔABC P.T.

(i) $\sin(B+C) = \sin A$

(ii) $\tan\left(\frac{B+C}{2}\right) = \cot\frac{A}{2}$

(iii) $\sin\left(\frac{B+C}{2}\right) = \cos\frac{A}{2}$

* Compound angle

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$\sin(A-B) = \sin A\cos B - \cos A\sin B$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\cos(A-B) = \cos A\cos B + \sin A\sin B$

$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

$\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A\tan B}$

* P.T. $\tan 50 = \tan 40^\circ + 2\tan 10^\circ$

* $\sin(A+B)\cdot\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

* P.T. $\dfrac{\sin(B-C)}{\sin B\sin C} + \dfrac{\sin(C-A)}{\sin C\sin A} + \dfrac{\sin(A-B)}{\sin A\sin B} = 0.$

* FOR ΔABC P.T. $\tan A + \tan B + \tan C = \tan A\cdot\tan B\cdot\tan C.$

* P.T. $\tan 5A - \tan 3A - \tan 2A = \tan 5A\tan 3A\tan 2A$

* P.T. $\tan 55 = \dfrac{\cos 10 + \sin 10}{\cos 10 - \sin 10}$

* $\tan 57 = \dfrac{\cos 12 + \sin 12}{\cos 12 - \sin 12}$

* P.T. $\tan 20^\circ + \tan 25^\circ + \tan 20^\circ\tan 25^\circ = 1.$

* $\tan 66^\circ =$

* P.T. $(1 + \tan 25^\circ)(1 + \tan 20^\circ) = 2$

* P.T. $\tan 10^\circ + \tan 35^\circ + \tan 10^\circ\tan 35^\circ = 1.$

$*$ $\sin 2\theta = 2\sin\theta\cos\theta$ $\qquad$ $*$ Multiple - submultiple angle

$\sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta$

$= 2\sin\theta\cos\theta$

$\rightarrow \sin 2\theta = \frac{2\sin\theta\cos\theta}{1} = \frac{2\sin\theta\cos\theta}{\sin^2\theta+\cos^2\theta} = \frac{2\frac{\sin\theta}{\cos\theta}}{\tan^2\theta+1}$

$\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

$*$ $\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\cos 2\theta = \cos(\theta+\theta) = \cos\theta\cos\theta - \sin\theta\sin\theta$

$= \cos^2\theta - \sin^2\theta$

$\rightarrow \cos 2\theta = \cos^2\theta - (1-\cos^2\theta) = 2\cos^2\theta - 1$

$\rightarrow \cos 2\theta = 1-\sin^2\theta - \sin^2\theta = 1-2\sin^2\theta$

$\rightarrow \cos 2\theta = \frac{\cos^2\theta - \sin^2\theta}{1} = \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta+\sin^2\theta} = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

dividing $\cos^2\theta$

$*$ $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

$\rightarrow \tan 2\theta = \tan(\theta+\theta) = \frac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta} = \frac{2\tan\theta}{1-\tan^2\theta}$

$*$ $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

$\sin 3\theta = \sin(2\theta+\theta)$

$= \sin 2\theta\cos\theta + \cos 2\theta\sin\theta$

$= (2\sin\theta\cos\theta)\cos\theta + (1-2\sin^2\theta)\cdot\sin\theta$

$= 2\sin\theta\cos^2\theta + \sin\theta - 2\sin^3\theta$

$\therefore \sin 3\theta = 2\sin\theta(1-\sin^2\theta) + \sin\theta - 2\sin^3\theta$

$= 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta$

$= 3\sin\theta - 4\sin^3\theta$

* $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

L.H.S $= \cos 3\theta = \cos(2\theta + \theta)$

$= \cos 2\theta \cos\theta - \sin 2\theta \sin\theta$

$= (2\cos^2\theta - 1)\cos\theta - (2\sin\theta\cos\theta)\sin\theta$

$= 2\cos^3\theta - \cos\theta - 2\sin^2\theta\cos\theta$

$= 2\cos^3\theta - \cos\theta - 2(1-\cos^2\theta).\cos\theta$

$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta.$

$=$ R.H.S

* $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

* $\frac{\theta}{2}$ Formula

* P.T. $\dfrac{1+\sin 2A - \cos 2A}{1+\sin 2A + \cos 2A} = \tan A$, $\dfrac{1+\sin\theta - \cos\theta}{1+\sin\theta + \cos\theta} = \tan\frac{\theta}{2}$

* P.T. $\dfrac{\sin A + \sin 2A}{1+\cos A + \cos 2A} = \tan A$

$\dfrac{\sin A + 2\sin A\cos A}{1+\cos A + 2\cos^2 A - 1}$

$\dfrac{\sin A(1+2\cos A)}{\cos A(1+2\cos A)} = \tan A$

* If $\tan\theta = -\frac{3}{4}$ then find $\sin 2\theta$, $\cos 2\theta$.

* $S + S = 2SC$

$S - S = 2CS$

$C + C = 2CC$

$C - C = -2SS$

* $2\sin A \cos B = \sin(A+B) + \sin(A-B)$.

$2\cos A \sin B = \sin(A+B) - \sin(A-B)$

$2\cos A \cos B = \cos(A+B) + \cos(A-B)$

$-2\sin A \sin B = +\cos(A+B) - \cos(A-B)$

* $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right)$

$\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right).\sin\left(\frac{A-B}{2}\right)$

$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$.

* P.T. $\dfrac{\sin\theta + \sin 2\theta + \sin 3\theta}{\cos\theta + \cos 2\theta + \cos 3\theta} = \tan 2\theta$

* P.T. $\dfrac{\cos A + \cos 3A + \cos 5A}{\sin A + \sin 3A + \sin 5A} = \cot 3A$

* P.T. $\dfrac{\cos 4\theta + 2\cos 5\theta + \cos 6\theta}{\sin 4\theta + 2\sin 5\theta + \sin 6\theta} = \cot 5\theta$

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* Inverse Trigonometry Function

* P.T. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

Let $\sin^{-1}x = \theta$

$\therefore x = \sin\theta$

we know that $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

$\therefore x = \cos\left(\frac{\pi}{2} - \theta\right)$

$\therefore \cos^{-1}x = \frac{\pi}{2} - \theta$

$\therefore \cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$

* P.T. $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$

* P.T. $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$.

* $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $xy < 1$

$= \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $xy > 1$

$= \frac{\pi}{2}$, if $xy = 1$

* P.T. $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$

* P.T. $\tan^{-1}\left(\frac{5}{7}\right) + \tan^{-1}\left(\frac{1}{6}\right) = \frac{\pi}{4}$

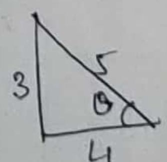
* P.T. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

* P.T. ✗ $\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

* P.T. $2\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$

* P.T. $2\tan^{-1}\left(\frac{2}{3}\right) = \tan^{-1}\left(\frac{12}{5}\right)$

* P.T. $\sin^{-1}\left(\frac{3}{5}\right) + \tan^{-1}\left(\frac{4}{3}\right) = \frac{\pi}{2}$.

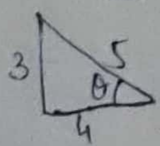


$\sin^{-1}\left(\frac{3}{5}\right) = \theta$

$\sin\theta = \frac{3}{5}$

$\tan\theta = \frac{3}{4}$

* P.T. $\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{4}{3}\right) = \frac{\pi}{2}$



$\cos\theta = \frac{4}{5}$

$\tan\theta = \frac{3}{4}$

* $y = \sin x$, $0 \le x \le 2\pi$ $y = \sin x$, $0 \le x \le \pi$

* $y = \cos x$, $0 \le x \le 2\pi$ $y = \cos x$, $0 \le x \le \pi$

* $y = \sin x$, $-\pi/2 \le x \le \pi/2$

* $y = \cos x$, $-\pi/2 \le x \le \pi/2$

* $y = 2\sin x$, $0 \le x \le \pi$

* $y = 2\cos x$, $0 \le x \le \pi$

* $y = \sin \frac{x}{2}$, $0 \le x \le 2\pi$

* $y = \cos \frac{x}{2}$, $0 \le x \le 2\pi$

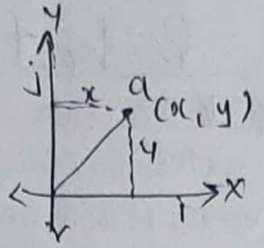
Unit-3 Vectors - 14 Marks

* Vector → Direction + Value (Displacement, flow of fluid)

  Scalar → only value. (Mass, work, Energy, Frequency, Electric charge)

* Co-ordinate form

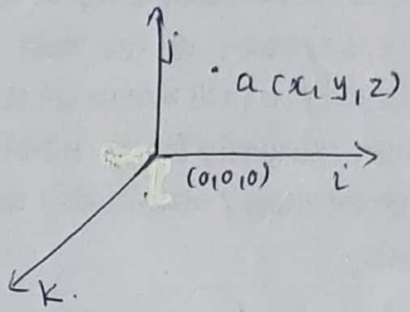
  $a(x, y)$



  vector form

  $\bar{a} = xi + yj$

* $a(x, y, z) \Rightarrow \bar{a} = xi + yj + zk$



* $a(1, -1, 3) \Rightarrow \bar{a} = i - j + 3k$

  $b(5, 2, -3) \Rightarrow \bar{b} = 5i + 2j - 3k$

* $\bar{a} = 3i - j + 5k \Rightarrow a(3, -1, 5)$

* $\bar{b} = 4i - k \Rightarrow b(4, 0, -1)$

* $\bar{c} = 3i - 2k + j \Rightarrow c(3, 1, -2)$

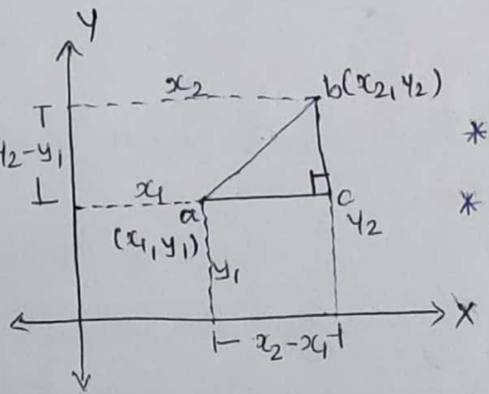
* Modulus

  $|-3| = 3$

  $|-4| = 4$

  $ab^2 = ac^2 + bc^2$

  $= (x_2 - x_1)^2 + (y_2 - y_1)^2$

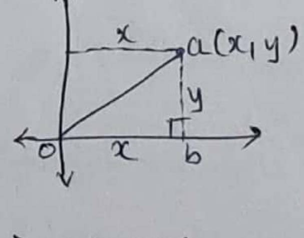


* $|\bar{x}| \geq 0$
* $|\bar{x}| = 0 \Leftrightarrow \bar{x} = \bar{0}$

→ $oa^2 = ab^2 + ob^2$

  $= x^2 + y^2$

  $\therefore oa = \sqrt{x^2 + y^2}$.



→ $a(x, y, z) \Rightarrow |\bar{a}| = \sqrt{x^2 + y^2 + z^2}$

* If $\bar{a} = i - 2j + k$ then $|\bar{a}| =$ ___

* $|3i - 4j + 5k| =$ ___

* If $\bar{a} = 2i - j + 4k$ then $|\bar{a}| =$ ___

* $|2i + 4j + 4k| =$ ___

* If $\bar{b} = i - 3k + j$ then $|\bar{b}| =$ ___

* If $\bar{c} = (1, 0, 0)$ then $|\bar{c}| =$ ___

* If $\bar{a}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ then $|\bar{a}| =$ ___

* If $\bar{x} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ then $|\bar{x}| =$ ____

* If $\bar{c} = (\sin\theta, \cos\theta)$ then $|\bar{c}| =$ ____

* **Unit Vector** :- If Modulus of vector is unit then given vector is called unit vector. $|\bar{a}| = 1$.

* **Unit vector of $\bar{a}$** :- $\hat{a} = \frac{\bar{a}}{|\bar{a}|}$

* If $\bar{a} = i - 3k + j$ then find unit vector of $\bar{a}$.

$$\hat{a} = \left(\frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{-3}{\sqrt{11}}\right)$$

→ For verification :- $\sqrt{x^2+y^2+z^2} = 1$.

* If $\bar{x} = 3i - j + 4k$ then find unit vector of $\bar{x}$.

$$\hat{x} = \left(\frac{3}{\sqrt{26}}, \frac{-1}{\sqrt{26}}, \frac{4}{\sqrt{26}}\right)$$

* **Addition & substraction of vector**

→ $\bar{a} = (x_1, y_1, z_1)$ $\bar{b} = (x_2, y_2, z_2)$

$\bar{a} \pm \bar{b} = (x_1, y_1, z_1) \pm (x_2, y_2, z_2)$

$= (x_1 \pm x_2, y_1 \pm y_2, z_1 \pm z_2)$

* If $\bar{a} = i - 3j + 5k$ and $\bar{b} = 4i + j - 2k$ then find (i) $\bar{a} + \bar{b}$ (ii) $\bar{a} - \bar{b}$ (iii) $2\bar{a} + 3\bar{b}$ (iv) $3\bar{a} - 2\bar{b}$.

→ (i) $\bar{a} + \bar{b} = (5, -2, 3)$ (iii) $2\bar{a} + 3\bar{b} = (14, -3, 4)$

(ii) $\bar{a} - \bar{b} = (-3, -4, 7)$ (iv) $3\bar{a} - 2\bar{b} = (-5, -11, 19)$

* If $\bar{a} = j + k - i$ and $\bar{b} = 2i + j - 3k$ then find the value of $|2\bar{a} + 3\bar{b}|$. $= 3\sqrt{10}$

* If $\bar{a} = (3, -1, -4)$, $\bar{b} = (-2, 4, -3)$ and $\bar{c} = (-1, 2, -5)$ then find $|\bar{a} + 2\bar{b} - \bar{c}|$. $= \sqrt{50}$

* If $\bar{a} = i - 2j + k$, $\bar{b} = 2i + j + 3k$ and $\bar{c} = -i + 2j - 3k$ then find $|2\bar{a} - 3\bar{b} + \bar{c}|$. $= 5\sqrt{6} = \sqrt{150}$

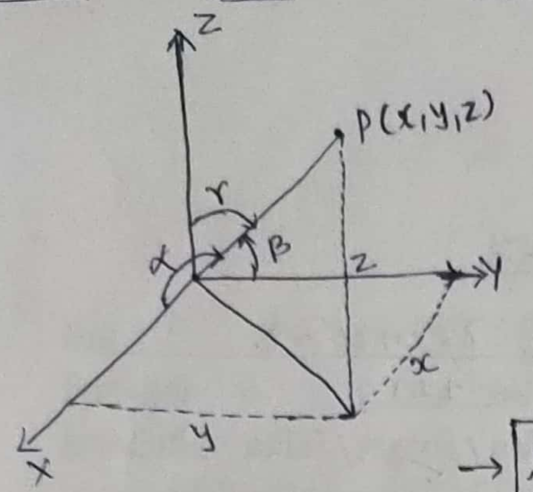
* If $\bar{a} = (3, -1, -4)$, $\bar{b} = (-2, 4, -3)$ and $\bar{c} = (-1, 2, -1)$ then find $|3\bar{a} - 2\bar{b} + 4\bar{c}|$. $= \sqrt{190}$

* If $\bar{a} = i - 2j + 4k$, $\bar{b} = -3i + j - 4k$ and $\bar{c} = i + 2j - 4k$ then find $|5\bar{a} + 3\bar{b} + 2\bar{c}|$. $= \sqrt{130}$

* If $\bar{a} = i + 2j + k$, $\bar{b} = 2i - 3j + k$ and $\bar{c} = -2i - j + 5k$ then find $|2\bar{a} + 3\bar{b} - \bar{c}|$.

* If $\bar{a} = 3i - 2j + k$, $\bar{b} = 2i - 4j - 3k$ and $\bar{c} = -i + 2j + 2k$ then find $|2\bar{a} - 3\bar{b} - 5\bar{c}| = \sqrt{30}$

* Direction cosine of vector :-



$l = \cos\alpha = \dfrac{x}{\sqrt{x^2+y^2+z^2}}$

$m = \cos\beta = \dfrac{y}{\sqrt{x^2+y^2+z^2}}$

$n = \cos\gamma = \dfrac{z}{\sqrt{x^2+y^2+z^2}}$.

$\rightarrow \boxed{l^2 + m^2 + n^2 = 1}$

* If $\bar{a} = 3i - j - 4k$, $\bar{b} = -2i + 4j - 3k$ and $\bar{c} = i + 2j - k$ then find the direction cosine of vector $3\bar{a} - 2\bar{b} + 4\bar{c}$.

* If $\bar{a} = 3i - j - 4k$, $\bar{b} = -2i + 4j - 3k$ and $\bar{c} = -i + 2j - 5k$ then find the direction cosine of $\bar{a} + 2\bar{b} - \bar{c}$.

* Multiplication of vectors

* Dot product $(\bar{a}\cdot\bar{b})$ Scalar product
* Cross product $(\bar{a}\times\bar{b})$ Vector product
* Dot product $\bar{a} = (x_1, y_1, z_1)$, $\bar{b} = (x_2, y_2, z_2)$

$\bar{a}\cdot\bar{b} = (x_1, y_1, z_1)\cdot(x_2, y_2, z_2)$
$= x_1x_2 + y_1y_2 + z_1z_2$.

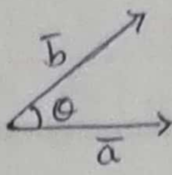
* If $\bar{a} = i - 2j + k$ and $\bar{b} = 3i + j + 4k$ then find (i) $\bar{a}\cdot\bar{b}$ (ii) $\bar{a}\cdot(\bar{a}+\bar{b})$. (iii) $\bar{a}\cdot(\bar{a}-\bar{b})$.

* If $\bar{x} = (1,-2,3)$ and $\bar{y} = (-2,3,1)$ then find $(\bar{x}+\bar{y})\cdot(\bar{x}-\bar{y})$.
* If $\bar{x} = (1,-2,3)$ and $\bar{y} = (1,2,-2)$ then find $(\bar{x}+\bar{y})\cdot(\bar{x}-\bar{y})$.
* If $\bar{a} = (-4,9,6)$, $\bar{b} = (0,7,10)$ and $\bar{c} = (-1,6,6)$ then P.T. $(\bar{a}-\bar{c})\cdot(\bar{b}-\bar{c}) = 0$.

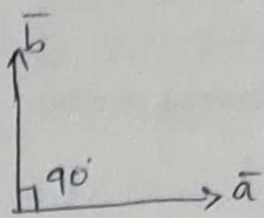
* $\bar{x}\cdot\bar{x} \geq 0$
* $\bar{x}\cdot\bar{x} = 0 \Leftrightarrow \bar{x} = \bar{0}$
* $\bar{x}\cdot\bar{y} = \bar{y}\cdot\bar{x}$
* $\bar{x}\cdot(\bar{y}+\bar{z}) = \bar{x}\cdot\bar{y} + \bar{x}\cdot\bar{z}$

* Angle between two Vectors

$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cdot|\bar{b}|}$$



$$\therefore \theta = \cos^{-1}\left(\frac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cdot|\bar{b}|}\right)$$



→ If $\bar{a}\cdot\bar{b} = 0$ then $\cos\theta = 0$
$\theta = \cos^{-1}0 = 90°$

* If $\bar{a}\cdot\bar{b} = 0$ then $\bar{a}$ and $\bar{b}$ are perpendicular to each other.
* If $\bar{a}$ & $\bar{b}$ are perpendicular to each other then $\bar{a}\cdot\bar{b} = 0$.
* $\sin^2\theta + \cos^2\theta = 1$

* If $\bar{x} = (1,1,1)$ and $\bar{y} = (2,-1,-1)$ then P.T. $\bar{x}$ and $\bar{y}$ are perpendicular to each other.
* If $\bar{x} = (1,-2,-3)$ and $\bar{y} = (2,P,4)$ then For what value of 'P'. $\bar{x}$ and $\bar{y}$ are perpendicular to each other. $(P = -5)$.
* Find x, If $\bar{a} = (2,-3,5)$ and $\bar{b} = (x,-6,-8)$ are perpendicular to each other. $(x = 11)$.
* If $2i+3j+k$ and $pi-j-3k$ are perpendicular to each other then find 'p'. $(p = 3)$.
* If $(m, 2m, 4)$ and $(m,-3,2)$ are perpendicular to each other then find m. $(m = 4$ or $m = 2)$

* Find the angle between two Vectors $(1,2,3)$ and $(-2,3,1)$.
* Find the angle between vectors $(1,2,4)$ and $(3,1,2)$.
* prove that the angle between two vectors $i+j-k$ and $2i-2j+k$ is $\sin^{-1}\sqrt{\frac{26}{27}}$.
* show that the angle between two vectors $i+2j$ and $i+j+3k$ is $\sin^{-1}\sqrt{\frac{46}{55}}$
* prove that the angle between two vectors $i+2j-3k$ and $2i+j-k$ is $\sin^{-1}\sqrt{\frac{35}{84}}$.
* P.T. the angle between two vectors $3i+j+2k$ and $2i-2j+4k$ is $\sin^{-1}\frac{2}{\sqrt{7}}$

* Cross product $(\bar{a} \times \bar{b})$

$\bar{a} = (x_1, y_1, z_1)$, $\bar{b} = (x_2, y_2, z_2)$

* $\bar{x} \times \bar{y} = -(\bar{y} \times \bar{x})$
* $\bar{x} \times \bar{x} = \bar{0}$
* $\bar{x} \times (\bar{y} + \bar{z}) = \bar{x} \times \bar{y} + \bar{x} \times \bar{z}$

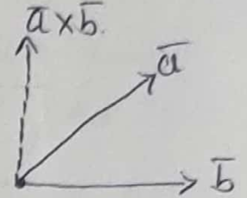
* $|i| = |j| = |k| = 1$
* $i \cdot i = j \cdot j = k \cdot k = 1$
* $i \cdot j = j \cdot k = k \cdot i = 0$

* $i \times j = k$, $j \times k = i$, $k \times i = j$
* $i \times i = 0$, $j \times j = 0$, $k \times k = 0$

$$\bar{a} \times \bar{b} = \begin{vmatrix} i & j & k \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = i \begin{vmatrix} y_1 & z_1 \\ y_2 & z_2 \end{vmatrix} - j \begin{vmatrix} x_1 & z_1 \\ x_2 & z_2 \end{vmatrix} + k \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

* If $\bar{a} = i - j + 3k$ and $\bar{b} = 3i + 2j - k$ then find (i) $\bar{a} \times \bar{b}$ (ii) $\bar{b} \times \bar{a}$.
* If $\bar{a} = 2i - j$ and $\bar{b} = i + 3j - 2k$ then find (i) $|\bar{a} \times \bar{b}|$ (ii) $|(\bar{a} + \bar{b}) \times (\bar{a} - \bar{b})|$.
* If $\bar{a} = (2, -3, -1)$ and $\bar{b} = (1, 4, -3)$ then find $|(\bar{a} + \bar{b}) \times (\bar{a} - \bar{b})|$.
* Simplify. $(10i + 2j + 3k) \cdot [(i - 2j + 2k) \times (3i - 2j - 2k)]$

Box product $\bar{a} \cdot (\bar{b} \times \bar{c}) = [\bar{a}\ \bar{b}\ \bar{c}] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$



* Unit vector perpendicular to both vectors :-

U. V. P. to both $\bar{a}$ & $\bar{b}$ $= \pm \dfrac{\bar{a} \times \bar{b}}{|\bar{a} \times \bar{b}|}$ $\begin{cases} \bar{a} \perp (\bar{a} \times \bar{b}) \\ \bar{b} \perp (\bar{a} \times \bar{b}). \end{cases}$

* If $\bar{x} = 3i - j + 2k$ and $\bar{y} = 2i + j - k$ then find unit vector perpendicular to both vectors $\bar{x}$ and $\bar{y}$. $\dfrac{(-1, 7, 5)}{\sqrt{75}}$
* Find the unit vector perpendicular to both vectors $\bar{a} = (5, 7, -2)$ and $\bar{b} = (3, 1, -2)$. $\dfrac{(-12, 4, -16)}{\sqrt{416}}$.
* Find the unit vector perpendicular to both vectors $\bar{a} = (1, 2, 3)$ and $\bar{b} = (-2, 1, -2)$. $\dfrac{(-7, -4, 5)}{\sqrt{90}}$
* Find the unit vector perpendicular to both vectors $\bar{a} = (3, 1, 2)$ and $\bar{b} = (2, -2, 4)$
* If $\bar{a} = 2i - 3j + 4k$ and $\bar{b} = i - j + k$ then find unit vector perpendicular to $\bar{a} + \bar{b}$ and $\bar{a} - \bar{b}$. $\dfrac{(-2, -4, -2)}{\sqrt{24}}$
* If $\bar{x} = (1, 1, 1)$ and $\bar{y} = (2, -1, -1)$ then P.T. $\bar{x}$ is perpendicular to $\bar{y}$. Also find unit vector perpendicular to both $\bar{x}$ and $\bar{y}$.

* Work done by force :- (W)

$\boxed{W = F \cdot d}$, where F = Total force $F_1 + F_2 + F_3 + \ldots$

d = Displacement $d_2 - d_1$

* The forces $3i - 2j + k$ and $-i - j + 2k$ act on a particle and particle moves from the point $(2, 2, -3)$ to the point $(-1, 2, 4)$ under the effect of these forces. Find the work done. $(W = 15 \text{ Units})$

* The constant forces (1,2,3) and (3,1,2) act on a particle and particle moves from the point (0,1,-2) to the point (5,1,2). Find the work done. (w 40 units)

* A particle moves from (-1,2,1) to (2,3,-1) under the effect of the forces (1,2,1) and (2,-1,0). Find work done. (w = 8 Units)

* The constant forces (1,2,3), (-1,2,3) and (-1,2,-3) act on a particle and under the effect of these forces particle move to the point (-1,3,2) from the point (0,1,-2). Find work done. (w = 25 Units).

* The constant forces (1,2,3) and (3,1,1) act on a particle and particle moves from the point (0,1,-2) to the point (5,1,2). Find the work done. w = 36 unit.

* The constant forces (1,-1,1) (1,1,-3) and (4,5,-6) act on a particle and particle moves from the point (3,-2,1) to the point (1,3,-4). Find work done. (w = 53 Units)

* <u>Moment of force</u> :-

→ The moment of a force $\bar{F}$ about a point $A(\bar{a})$ is a vector which is given by $\vec{AP} \times \bar{F}$, where $P(\bar{p})$ is a point on the line of force $\bar{F}$.

→ Magnitude of the moment of a force $\bar{F}$ about a point $A(\bar{a})$ on a particle $P(\bar{p}) = |\vec{AP} \times \bar{F}|$.

passing through the point / acting at the point } P. about the point } → A

→ $AP = OP - OA \xrightarrow{(\bar{p}-\bar{a})} \vec{AP} \times \bar{F}$

* A force $\bar{F} = 2i + j + k$ is acting at the point $\overset{P}{(-3,2,1)}$. Find the magnitude of the moment of force about the point $(2,1,2)^{A}$.
[$\vec{AP} \times \bar{F} = (2,3,-7)$, $|\vec{AP} \times \bar{F}| = \sqrt{62}$]

* A force $3i - j + 2k$ is acting at the point (1,2,-1). Find the moment of force about the point (3,0,1)
[$\vec{AP} \times \bar{F} = (2, -2, -4)$

Unit-4 Co-ordinate Geometry - 14 Marks

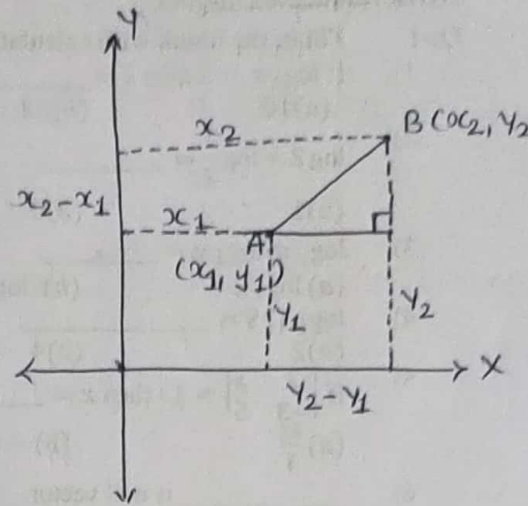
* **Distance Formula**

If $A(x_1, y_1)$ and $B(x_2, y_2)$ then

$$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$AB^2 = (x_2-x_1)^2+(y_2-y_1)^2$$

$$d(AB) = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$



* Find the distance between the points (4,5) and (7,3).
* Find the distance between the points (1,3) and (0,-4).
* Find the distance between the points (1,1) and (2,-1).
* Find the distance between the points (7,-5) and (3,-2).
* Find the distance between the points (-1,2) and (-7,6).
* $d[(3,2), (-1,1)] =$ ______
* If p is the mid point of a line segment AB for the points A(-2,-1) and B(4,3) then find P.
* If the mid point of line segment AB is (1,1) and B(4,3) then find co-ordinate of A.
* If the distance between the point (5,7) and (-3,m) is 10 then find the value of m.

* **Co-linearity**

$$D = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

If D=0 then given three vertices are co-linear.

(3,2) (5,4) (7,6)

(1,0) (0,1) (-1,2)

→ Using Distance Formula

(-3,-2) (5,2) (9,4)

AB = AC + BC

AC = AB + BC

BC = AB + AC

* Prove that $(3,2)$, $(5,4)$ and $(7,6)$ are co-linear.

* prove that $(1,0)$, $(0,1)$ and $(-1,2)$ are co-linear.

* prove that $(-3,-2)$, $(5,2)$ and $(9,4)$ are co-linear.

* prove that $(a, b+c)$, $(b, c+a)$, and $(c, a+b)$ are co-linear.

* If three point $(-k,1)$, $(k,3)$ and $(6,5)$ are co-linear then find value of k.

* Generalized equation of line $ax+by+c=0$

→ x-intercept $= -\frac{c}{a}$

→ y-intercept $= -\frac{c}{b}$

* To find slope (m)

→ Given two points (x_1, y_1) & (x_2, y_2).

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

→ Given angle θ then $m = \tan\theta$

→ Given line $ax+by+c=0$ then $m = -\frac{a}{b}$.

* Find the slope of line passing through the point (3, 2) and (1, 4).
* Find the slope of line passing through the point (8, 5) and (1, -2).
* Find the slope of line passing through the point (1, 3) and (4, -5).
* Find slope and intercepts of the line $4x+3y-7=0$.
* Find slope and intercepts of the line (i) $2x-3y+5=0$

(ii) $3x+5=0$

(iii) $2y-3x+4=0$

* Find slope and intercepts of line (i) $2x+y-8=0$

(ii) $2x-5y+3=0$ (iii) $2x+3y-4=0$

* Find slope of a line $(\cos\alpha)x + (\sin\alpha)y = 5$.
* To find Equation of line :-

→ Given two points (x_1, y_1) & (x_2, y_2) then eq^n $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0.$

* (2, 1) (-1, 3) * (2, 3) (3, -1) * (1, 6) (-2, 5)

$2x+3y-7=0$ $4x+y-11=0$

→ Given point (x_1, y_1) and slope (m) then eq^n (-2, -3), 3/2

$y - y_1 = m(x - x_1)$ (point-slope equation)

→ Given slope (m) and Intercept (c) on Y-axis, then eq^n

$y = mx + c$.

* Find the eq^n of line passing through the points $(2,1)$ & $(-1,3)$.

$(2x+3y-7=0)$

* Find the eq^n of line passing through the points $(2,3)$ & $(3,-1)$.

$(4x+y-11=0)$

* Find the eq^n of line passing through the points $(1,6)$ & $(-2,5)$.
* Find the eq^n of line passing through the point $(-2,-3)$ and having slope $3/2$.

$(3x-2y=0)$

* Find the eq^n of line passing through the point $(2,5)$ and having slope $-1/2$.
*

* **Angle between two lines:-**

$$\tan\theta = \left|\frac{m_1 - m_2}{1+m_1 m_2}\right|$$

$$\therefore \theta = \tan^{-1}\left|\frac{m_1 - m_2}{1+m_1 m_2}\right|,$$

where m_1 = slope of first line
m_2 = slope of second line
θ = Angle between two lines.

* The angle between two line is 45°. If the slope of first line is $\frac{2}{3}$. Find slope of other line. $(m_2 = 5 \text{ or } -1/5)$
* Find the angle between two straight line $x+y=0$ and $x-y=0$. $(\theta = \pi/2)$
* Find the angle between two straight line $\sqrt{3}x-y+1=0$ and $x-\sqrt{3}y+2=0$. $(\theta = \frac{\pi}{6})$
* Find the angle between two straight line $x+y+1=0$ and $2x+3y+7=0$. $(\theta = \frac{\pi}{4})$

* Condition of two parallel line $m_1 = m_2$
* Condition of two perpendicular line $m_1 \cdot m_2 = -1$.

* Prove that line $3x+2y+1=0$ and $6x+4y+3=0$ are parallel.
* Prove that line $7x+y-1=0$ and $3x-21y+2=0$ are perpendicular.
* Check the line $3x+2y+1=0$ and $2x-3y+7=0$ are parallel or perpendicular.
* If two lines $3x+4my+8=0$ and $3my-9x+10=0$ are perpendicular to each other then find value of m.
* If two lines $5x-py=3$ and $2x+3y=4$ are perpendicular to each other then find value of p.
  parallel then find P
* If two lines $3mx-2my-10=0$ and $(5m+2)x-4my-28=0$ are parallel to each other then find value of m. $(m=2)$

* General eq^n of line parallel to line $ax+by+c=0$ is $ax+by+k=0$
* General eq^n of line perpendicular to line $ax+by+c=0$ is
  $bx-ay+k=0$

+	+	−
−	−	−
+	−	+
−	+	+

* Find the equation of line parallel to the line $2x+y-1=0$ and passing through the point $(4,5)$. $(2x+y-13=0)$
* Find the equation of line parallel to the line $4x-3y+7=0$ and passing through the point $(4,3)$. $(4x-3y-7=0)$
* Find the equation of line perpendicular to the line $4x-y+5=0$ and passing through the point $(1,-2)$. $(x+4y+7=0)$
* Find the equation of line perpendicular to the line $x-3y+3=0$ and passing through the point $(-1,2)$. $(3x+y+1=0)$
* Find the equation of line perpendicular to the line $-3x+4y+7=0$ and passing through the point $(4,3)$. $(4x+3y-25=0)$
* Find the equation of line perpendicular to the line $3x-2y+4=0$ and passing through the point $(1,3)$. $(2x+3y-11=0)$
* Find the equation of line perpendicular to the line $y-4x+1=0$ and passing through the point $(2,1)$. $(x+4y-6=0)$

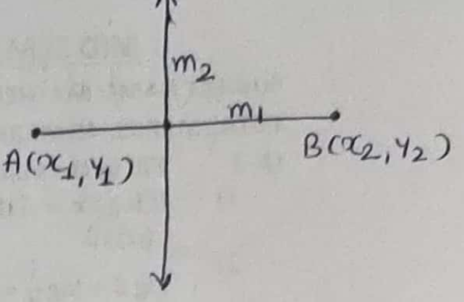
* perpendicular Bisector

→ Mid point $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

→ slope $\frac{y_2-y_1}{x_2-x_1}$

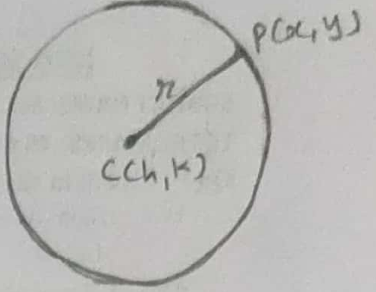
→ $\perp^{lar}$ line slope

→ point-slope equation $y-y_1=m(x-x_1)$.



* Find the equation of line which is perpendicular bisector of line joining points $(8,-2)$ and $(6,4)$. $(x-3y-4=0)$
* Find the equation of line which is perpendicular bisector of line joining points $(3,5)$ and $(1,1)$ $(x+2y-8=0)$
* Find the equation of line which is perpendicular bisector of line joining points $(-1,2)$ and $(1,-2)$. $(x-2y=0)$
* Find the equation of line which is perpendicular bisector of line joining points $(5,6)$ and $(-1,1)$. $(12x+10y-59=0)$.

CIRCLE



$CP = r$

$\therefore CP^2 = r^2$

$\therefore (x-h)^2 + (y-k)^2 = r^2$

centre point $C(h,k)$ and radius r then eq^n of circle is

$$(x-h)^2 + (y-k)^2 = r^2$$

→ If centre point $C(0,0)$ and radius r then eq^n of circle is

$$x^2 + y^2 = r^2$$

* Find equation of circle having centre $(6,7)$ and radius 5.
  $(x^2+y^2-12x-14y+60=0)$
* Find equation of circle having centre $(-2,5)$ and radius 4.
  $(x^2+y^2+4x-10y+13=0)$
* Find the equation of circle having centre $(-4,3)$ and tangent to x-axis. $(r=3)$
  $(x^2+y^2+4x-6y+4=0)$
* Find the equation of circle having centre $(3,-4)$ and tangent to y-axis. $(r=3)$
  $(x^2+y^2-6x+8y+16=0)$
* Find the equation of circle having centre $(1,1)$ and passing through $(-2,4)$.
  $(x^2+y^2-2x-2y-16=0)$
* Find the equation of circle having centre $(4,3)$ and passing through $(7,-2)$.
  $(x^2+y^2-8x-6y-9=0)$
* Find the equation of circle having centre $(2,3)$ and passing through $(3,4)$.
  $(x^2+y^2-4x-6y+11=0)$
* Find the equation of circle having centre $(3,4)$ and passing through origin.
  $(x^2+y^2-6x-8y=0)$
* centre $(-2,5)$ passing through the intersection of lines $2x+y-3=0$ $x-3y+2=0$

  $x=1, y=1$, $r^2=25$ $x^2+y^2+4x-12y+4=0$.

* General equation of circle

$x^2+y^2+2gx+2fy+c=0$.

centre $= (-g, -f)$ radius $r=\sqrt{g^2+f^2-c}$

$\rightarrow$ $x^2+y^2+2gx+2fy+c=0$

$\therefore x^2+2gx+g^2+y^2+2fy+f^2+c-g^2-f^2=0$

$\therefore (x+g)^2+(y+f)^2=g^2+f^2-c$

$\therefore (x-(-g))^2+(y-(-f))^2=g^2+f^2-c$.

comparing with $(x-h)^2+(y-k)^2=r^2$

then we have $C(h,k)=C(-g,-f)$

$r^2=g^2+f^2-c$

* Find centre and radius of circle $x^2+y^2-2x+4y-1=0$.

(centre $(1,-2)$, $r=\sqrt{6}$)

* Find centre and radius of circle $x^2+y^2-4x-6y-4=0$

(centre $(2,3)$, $r=\sqrt{17}$)

* Find centre and radius of circle $4x^2+4y^2+8x-12y-3=0$

(centre $(-1,3/2)$, $r=2$)

* Find centre and radius of circle $2x^2+2y^2+4x+6y-7=0$

(centre $(-1,-3/2)$, $r=\sqrt{\frac{27}{4}}$)

* Find centre and radius of circle $36x^2+36y^2+24x-36y-23=0$.

(centre $(-1/3, 1/2)$, $r=1$)

* If radius of a circle $x^2+y^2-4x-8y+k=0$ is 4 unit. then find value of k. ($k=4$)

* If radius of a circle $2x^2+2y^2-4x-8y+k=0$ is 4 unit. then find value of k. ($k=-22$).

* If radius of a circle $x^2+y^2-4x-4y+k=0$ is 4 unit then find value of k. ($k=-8$).

* Find the eq^n of circle having centre $(-2,5)$ and passing through the intersection of lines $2x+y-3=0$ and $x-3y+2=0$.

($x=1, y=1$, $r^2=25$

$x^2+y^2+4x-10y+4=0$)

* Equation of tangent and Normal to the circle $x^2+y^2=r^2$

· Equation of tangent $xx_1+yy_1=r^2$

Equation of Normal $\frac{x}{x_1}=\frac{y}{y_1}$

* Equation of tangent and Normal to the circle $x^2+y^2+2gx+2fy+c=0$

→ Equation of tangent

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0.$$

→ Equation of Normal

$$\frac{x-x_1}{x_1+g}=\frac{y-y_1}{y_1+f}$$

* Find the eq^n of tangent and Normal to the circle $x^2+y^2=52$ at point $(-6,4)$. [Tangent $6x-4y+52=0$
$2x+3y=0$]

* Find the eq^n of tangent and Normal to the circle $x^2+y^2-6x+10y+21=0$ at point $(1,-2)$.
[Tangent $2x-3y-8=0$
Normal $3x+2y+1=0$]

* Find the eq^n of tangent and Normal to the circle $x^2+y^2-4x+2y+3=0$ at point $(1,-2)$.
[Tangent $x+y+1=0$, Normal $x-y-3=0$]

* Find the eq^n of tangent and Normal to the circle $x^2+y^2-2x+4y-20=0$ at point $(-2,2)$.
[Tangent $3x-4y+14=0$, Normal $4x+3y+2=0$]

* Find the eq^n of tangent and Normal to the circle $x^2+y^2-2x-7=0$ at point $(2,3)$.
[Tangent $x+y-5=0$, Normal $x-y+1=0$]

* Find the eq^n of tangent and Normal to the circle $2x^2+2y^2+3x-4y+1=0$ at point $(-1,2)$.
[Tangent $x-4y+9=0$, Normal $4x+y+2=0$]

Unit - 5. Limit - 12 Marks

* $f(x) = \frac{x^2+2}{x-2}$

$f(0) = -1$, $f(-1) = -1$, $f(1) = 3$, $f(2) = \infty$

* $\lim_{x \to 2} \frac{x^2+2}{x-2}$ $\quad x \to 2 \quad x \neq 2 \quad x-2 \neq 0.$

* $\lim_{x \to a} \frac{x^2+a}{x-a}$ $\quad x \to a \quad x \neq a \quad x-a \neq 0$

* $\lim_{x \to 1} (x^3 - 3x^2 + 5x - 6)$ (-3)

* $\lim_{x \to 0} \frac{x^2+3x+2}{5x+2}$ (1)

* $\lim_{x \to 1} \frac{x^2-4x+3}{x+1}$ (0)

* $\lim_{x \to -2} \frac{x^3+2x^2-x-1}{x-2}$ $\left(-\frac{1}{4}\right)$

* $\lim_{x \to 1} \frac{x^2+x+1}{x+1}$ $\left(\frac{3}{2}\right)$

* $\lim_{x \to 2} \frac{x^2-5x+6}{x-2}$ (-1)

* $\lim_{x \to 1} \frac{x^3-x^2+x-1}{x^2-1}$ (1)

* $\lim_{x \to 1} \frac{x^2-4x+3}{x^2+2x-3}$ $(-1/2)$

* $\lim_{x \to 3} \frac{x^2-x-6}{x^3-3x^2+x-3}$ $(1/2)$

* $\lim_{x \to 2} \frac{x^2-5x+6}{x^2-4}$ $\left(-\frac{1}{4}\right)$

* $\lim_{x \to 2} \frac{x^3-x^2-5x+6}{x^2-5x+6}$ (-3)

* $\lim_{x \to 1} \frac{x^2-6x+5}{2x^2-5x+3}$ (4)

* $\lim_{x \to 2} \frac{x^3-3x^2+5x-6}{x^3-8}$ $\left(\frac{5}{12}\right)$

* $\lim_{x \to 1} \frac{x^2-8x+7}{7x^2-6x-1}$ $\left(-\frac{3}{4}\right)$

* $\lim_{x \to 2} \frac{x^3-x^2-5x+6}{x^3-8}$ $\left(\frac{1}{4}\right)$

* $\lim_{x \to -3} \frac{x^3+27}{x^2+5x+6}$ (-27)

* $\lim_{x \to -1} \frac{2x^3+5x^2+4x+1}{3x^3+5x^2+x-1}$ $\left(\frac{1}{4}\right)$

* $\lim_{x\to 0} \frac{\sqrt{9+x}-3}{x}$ $\left(\frac{1}{6}\right)$

* $\lim_{x\to 0} \frac{\sqrt{1-x}-\sqrt{1+x}}{x}$ (-1)

* $\lim_{x\to 0} \frac{\sqrt{9-x}-3}{x}$ $\left(-\frac{1}{6}\right)$

* $\lim_{x\to a} \frac{\sqrt{2a-x}-\sqrt{x}}{a-x}$ $\left(\frac{1}{\sqrt{a}}\right)$

* $\lim_{x\to 0} \frac{\sqrt{25+x}-5}{x}$ $\left(\frac{1}{10}\right)$

* $\lim_{n\to\infty} \frac{1}{n} = 0$

* $\lim_{n\to\infty} \frac{6n^2-3n+5}{2n^2+4n-3}$ (3)

* $\lim_{n\to\infty} \frac{3n^3-4n^2-n-5}{2n^3+3n^2-2n+7}$ $\left(\frac{3}{2}\right)$

* $\lim_{x\to\infty} \frac{5x^2-3x+2}{6x^2+3x-1}$ $\left(\frac{5}{6}\right)$

* $\lim_{n\to\infty} \frac{4n^3-7n^2+5n-1}{8n^3+7n^2-4n+1}$ $\left(\frac{1}{2}\right)$

* $\lim_{x\to\infty} \frac{x(x+1)}{x^2+5x+6}$ (1)

* $\lim_{x\to\infty} \frac{x^2+8x+2}{x^3+x-4}$ (0)

* $\lim_{x\to\infty} \sqrt{x^2+x}-x$ $\left(\frac{1}{2}\right)$

* $\lim_{x\to\infty} \sqrt{x}\left(\sqrt{x+p}-\sqrt{x}\right)$ $\left(\frac{p}{2}\right)$

* $\lim_{n\to\infty} \sqrt{n^2+n+1}-n$ $\left(\frac{1}{2}\right)$

* $\sum n = 1+2+3+\ldots+n = \frac{n(n+1)}{2}$

* $\sum n^2 = 1^2+2^2+3^2+\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$

* $\sum n^3 = 1^3+2^3+3^3+\ldots+n^3 = \frac{n^2(n+1)^2}{4}$

* $\lim_{n\to\infty} \frac{\sum n^2}{n^3}$ $\left(\frac{1}{3}\right)$

* $\lim_{n\to\infty} \frac{1^2+2^2+3^2+\ldots+n^2}{n^3+1}$

* $\lim_{h\to 0} \frac{a^h-1}{h} = \log_e a$ $\lim_{h\to 0} \frac{e^h-1}{h} = \log_e e = 1$

* $\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$ $\lim_{x\to 0} (1+x)^{1/x} = e$

* $\lim_{x\to 0} \frac{5^x-1}{x} = \log_e 5$ * $\lim_{x\to 0} \frac{5^{2x}-1}{x} = 2\log_e 5$

* $\lim_{x\to 0} \frac{3^x-1}{x} = \log_e 3$ * $\lim_{x\to 0} \frac{3^{2x}-1}{x} = 2\log_e 3$

* $\lim_{x\to 0} \frac{7^x-1}{x} = \log_e 7$ * $\lim_{x\to 0} \frac{2^{3x}-1}{x} = \log_e 8$

* $\lim_{x\to 0} \frac{3^x-2^x}{x} = \log \frac{3}{2}$ * $\lim_{x\to 0} \frac{e^{3x}-1}{x} = 3$

* $\lim_{x\to 0} \frac{5^x-3^x}{x} = \log \frac{5}{3}$ * $\lim_{x\to 0} \frac{3^{2x}-2^{3x}}{x} = \log \frac{9}{8}$

* $\lim_{x\to 0} \frac{a^x-b^x}{x} = \log \frac{a}{b}$ * $\lim_{x\to 0} \frac{e^{ax}-e^{bx}}{x} = (a-b)$

* $\lim_{x\to 0} \frac{8^x-4^x-2^x+1}{x^2}$ $(\log_e 2 \cdot \log_e 4)$ * $\lim_{x\to 0} \frac{2(3^x)+3(2^x)-5}{x}$

$2\log_e 3 + 3\log_e 2$ $\log_e 72$

* $\lim_{x\to 0} \frac{15^x-5^x-3^x+1}{x^2}$ $(\log_e 3 \cdot \log_e 5)$ * $\lim_{x\to\infty} x\cdot\left(\sqrt[x]{5}-1\right)$ $(\log_e 5)$

* $\lim_{x\to\infty} \frac{a^{1/x}-1}{1/x} = \log_e a$

* $\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$

$\lim_{x\to 0} (1+x)^{1/x} = e$

* $\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^{4n}$ (e^4)

* $\lim_{x\to\infty} \left(1+\frac{2}{x}\right)^{2x}$ (e^4)

* $\lim_{x\to\infty} \left(1+\frac{5}{x}\right)^{2x}$ (e^{10})

* $\lim_{x\to\infty} \left(\frac{x+1}{x+2}\right)^{x}$ $\left(\frac{1}{e}\right)$

* $\lim_{x\to 0} \left(1+\frac{2x}{3}\right)^{5/x}$ $(e^{10/3})$

* $\lim_{x\to 0} \left(1+\frac{3x}{4}\right)^{5/x}$ $(e^{15/4})$

* $\lim_{x\to 0} \left(1+\frac{5x}{7}\right)^{2/x}$ $(e^{10/7})$

* $\lim_{x\to 0} \left(1+\frac{3x}{2}\right)^{4/x}$ (e^{6})

* $\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$

* $\lim_{x\to 2} \frac{x^3-8}{x-2}$ (12)

* $\lim_{x\to 2} \frac{x^4-16}{x^3-8}$ $\left(\frac{8}{3}\right)$

* $\lim_{x\to 5} \frac{x^3-125}{x^2-25}$ $\left(\frac{15}{2}\right)$

* $\lim_{x\to a} \frac{\sqrt{x}-\sqrt{a}}{x-a}$ $\left(\frac{1}{2\sqrt{a}}\right)$

* $\lim_{x\to 3} \frac{x^3-27}{\sqrt[3]{x}-\sqrt[3]{3}}$ $(81\cdot 3^{2/3})$

* $\lim_{x\to 2} \frac{x\sqrt{x}-2\sqrt{2}}{x-2}$ $\left(\frac{3}{\sqrt{2}}\right)$

* $\lim_{x\to -1} \frac{x^{2021}+1}{x^{2022}+1}$

* $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$

* $\lim_{\theta \to 0} \frac{\tan\theta}{\theta} = 1$

* $\lim_{x \to 0} \frac{\sin^{-1}x}{x} = 1$

* $\lim_{x \to 0} \frac{\tan^{-1}x}{x} = 1$

* $\lim_{\theta \to 0} \frac{\sin 5\theta}{\theta}$ (5)

* $\lim_{x \to 0} \frac{x}{\sin x}$ (1)

* $\lim_{\theta \to 0} \frac{\sin m\theta}{\sin n\theta}$ $\left(\frac{m}{n}\right)$

* $\lim_{x \to 0} \frac{\sin 3x}{x}$ (3)

* $\lim_{x \to 0} \frac{\sin 3x}{2x}$ $\left(\frac{3}{2}\right)$

* $\lim_{x \to 0} \frac{\tan 5x}{\sin 3x}$ $\left(\frac{5}{3}\right)$

* $\lim_{\theta \to 0} \frac{\theta}{\tan 3\theta}$ $\left(\frac{1}{3}\right)$

* $\lim_{\theta \to 0} \frac{\sin 2\theta}{\tan 3\theta}$ $\left(\frac{2}{3}\right)$

* $\lim_{x \to \frac{\pi}{2}} \frac{1-\sin x}{\cos^2 x}$ $\left(\frac{1}{2}\right)$

* $\lim_{x \to 0} \frac{3\sin x - \sin 3x}{x^3}$ (4)

* $\lim_{x \to 0} \frac{1-\cos x}{x^2}$ $\left(\frac{1}{2}\right)$

* $\lim_{x \to 0} \frac{x^2}{1-\cos x}$ (2)

* $\lim_{x \to 0} \frac{(1-\cos x)\sin x}{x^3}$ $\left(\frac{1}{2}\right)$

* $\lim_{x \to 0} \frac{(1-\cos x)\tan x}{x^3}$ $\left(\frac{1}{2}\right)$

* $\lim_{x \to 0} \frac{\operatorname{cosec} x - \cot x}{x}$ $\left(\frac{1}{2}\right)$

* $\lim_{x \to \frac{\pi}{4}} \frac{2-\sec^2 x}{1-\tan x}$ (2) $(\sec^2 x = 1+\tan^2 x)$

* $\lim_{h \to 0} \frac{a^h - 1}{h} = \log_e a$

* $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$

* $\lim_{x \to 0} \frac{a^x + \sin x - 1}{x}$ $(\log_e a + 1)$

* $\lim_{x \to 0} \frac{e^x + \sin 2x - 1}{x}$ (3)

* $\lim_{x \to 0} \frac{e^x + \sin x - 1}{x}$ (2)

* $\lim_{x \to 0} \frac{e^x - \sin x - 1}{x}$ (0)